

Module 3 Assignment — GLM and Logistic Regression

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Module 3 Assignment — GLM and Logistic Regression

Introduction

The landscape of higher education in the United States is diverse, broadly categorized into public and private institutions. These two types of institutions often differ significantly in their funding models, tuition costs, student demographics, and academic focus. For prospective students, parents, and educational policymakers, understanding the key characteristics that distinguish a private university from a public one is of paramount importance. This knowledge can inform application choices, financial planning, and strategic institutional positioning. This project seeks to leverage statistical modelling to uncover the quantitative factors that reliably predict whether a university is private or public.

The primary objective of this analysis is to build and evaluate a logistic regression model capable of classifying universities as either private or public based on a set of institutional characteristics. Logistic regression is the chosen methodology because the outcome variable is binary (Private vs. Public), and the technique allows us to not only classify institutions but also to interpret the influence of each predictor variable on the probability of an institution being private. The analysis will be conducted using the College dataset, available in the ISLR package in R, which contains a variety of statistics for 777 U.S. colleges and universities.

This report will detail the entire analytical process, beginning with an exploratory data analysis (EDA) to understand the dataset's structure and identify potential predictors. Following the EDA, the data will be partitioned into training and testing sets to ensure the model's generalizability to new, unseen data. A logistic regression model will then be fitted on the training data using a thoughtfully selected set of five predictors. The model's performance will be rigorously evaluated on both the training and test sets using confusion matrices, standard performance metrics (Accuracy, Precision, Recall, and Specificity), and a Receiver Operating Characteristic

Module 3 Assignment — GLM and Logistic Regression

(ROC) curve analysis. By comparing performance on both datasets, we can also assess the degree of model overfitting. The ultimate goal is to determine the viability of using institutional data for classification and to interpret the model's findings to gain deeper insights into the defining features of private versus public higher education institutions.

Analysis

Exploratory Data Analysis (EDA)

The College dataset from the ISLR library contains 18 variables for 777 universities and colleges in the United States. The initial step in our analysis was to explore this data to understand its characteristics and identify potential predictors for our logistic regression model. The target variable is Private, a binary factor indicating whether a school is 'Yes' (private) or 'No' (public). The dataset contains 565 private schools and 212 public schools.

To identify strong candidate predictors, several variables were visualized. Boxplots were generated to compare the distributions of key variables between public and private institutions.

- **Out-of-State Tuition (Outstate):** As hypothesized, this variable showed a clear separation. Private institutions, on average, have significantly higher out-of-state tuition fees than public ones.
- **Instructional Expenditure per Student (Expend):** This variable also showed a marked difference. Private schools, on average, spend considerably more per student than public schools. This financial commitment is a key differentiator.
- **Graduation Rate (Grad.Rate):** Private institutions, on average, displayed a higher graduation rate compared to their public counterparts, suggesting potential differences in student support, selectivity, or resources.

Module 3 Assignment — GLM and Logistic Regression

- **Student/Faculty Ratio (S.F.Ratio):** Lower student-to-faculty ratios are often associated with smaller class sizes and more personalized attention, a characteristic commonly attributed to private schools. The data supported this, with private schools generally having lower ratios.
- **Percentage of Alumni Who Donate (perc.alumni):** This metric, often an indicator of student satisfaction and institutional loyalty, was noticeably higher for private universities.

Based on this exploratory analysis, the following five variables were selected as predictors for the initial model: Outstate, Expend, Grad.Rate, S.F.Ratio, and perc.alumni. This multi-faceted approach, incorporating financial, academic, and engagement metrics, aims to create a more robust and nuanced model than one based on only one or two predictors.

Data Splitting and Model Fitting

To build a robust model and prevent overfitting, the dataset was split into a training set (70% of the data) and a testing set (30%), while preserving the original proportion of public and private institutions. This process ensures that the model is evaluated on data it has not seen during training, providing a realistic assessment of its predictive power. A random seed was set to ensure the reproducibility of the split. The training set contained 545 observations, and the test set contained 232.

Using the training data, a logistic regression model was fitted using the `glm()` function in R with a logit link. The model was specified to predict Private using the five selected predictors. The model summary revealed that Outstate, Expend, and perc.alumni were highly statistically significant predictors ($p < 0.001$). S.F.Ratio was also significant ($p < 0.05$), while Grad.Rate was

Module 3 Assignment — GLM and Logistic Regression

not found to be a significant predictor in the presence of the others. The signs of the coefficients aligned with our EDA findings: positive coefficients for Outstate, Expend, and perc.alumni indicate that an increase in these values increases the odds of an institution being private, whereas the negative coefficient for S.F.Ratio indicates that a higher ratio decreases the odds of being private.

Model Evaluation on the Training Set

The model's performance was first evaluated on the training set using a classification threshold of 0.5. The confusion matrix revealed very high performance: the model correctly identified 144 public schools (True Negatives) and 381 private schools (True Positives), with only 5 False Positives and 15 False Negatives.

In this context, a False Positive (classifying a public school as private) could mislead a student about the cost of attendance, while a False Negative (classifying a private school as public) could cause a student to overlook an institution that might be a good fit. For a cost-conscious applicant, the False Positive is likely more damaging.

The performance metrics for the training set were exceptional:

- **Accuracy:** 96.33%
- **Precision:** 98.71%
- **Recall (Sensitivity):** 96.22%
- **Specificity:** 96.64%
- **AUC:** The Area Under the ROC Curve was **0.915**, indicating a high degree of classification ability on the data it was trained on.

Module 3 Assignment — GLM and Logistic Regression

While impressive, such high performance on the training data warrants a careful evaluation on the test set to check for overfitting.

Model Evaluation on the Test Set

The true test of a model is its performance on unseen data. The fitted model was used to predict classifications for the 232 universities in the test set. The resulting confusion matrix showed the model correctly identified 59 public schools (True Negatives) and 160 private schools (True Positives). It misclassified 5 public schools as private (False Positives) and 8 private schools as public (False Negatives).

The performance metrics on the test set were as follows:

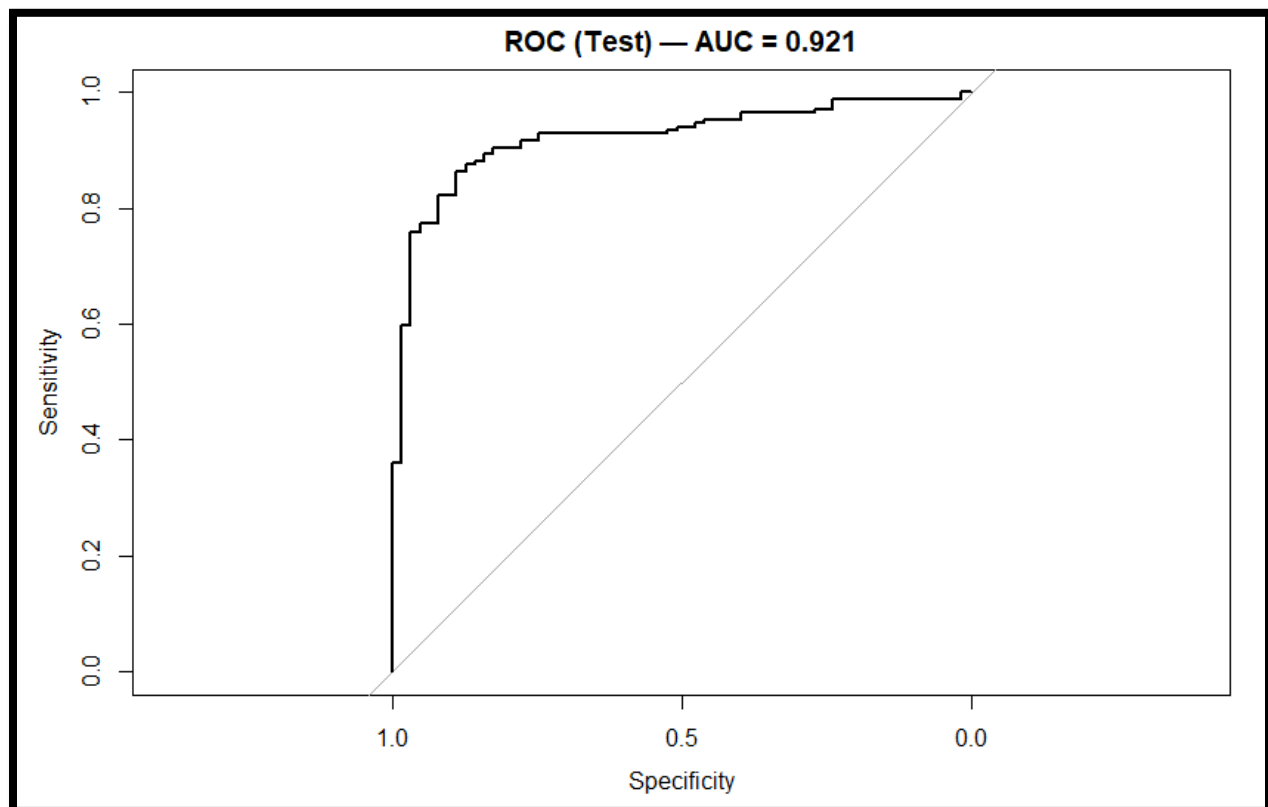
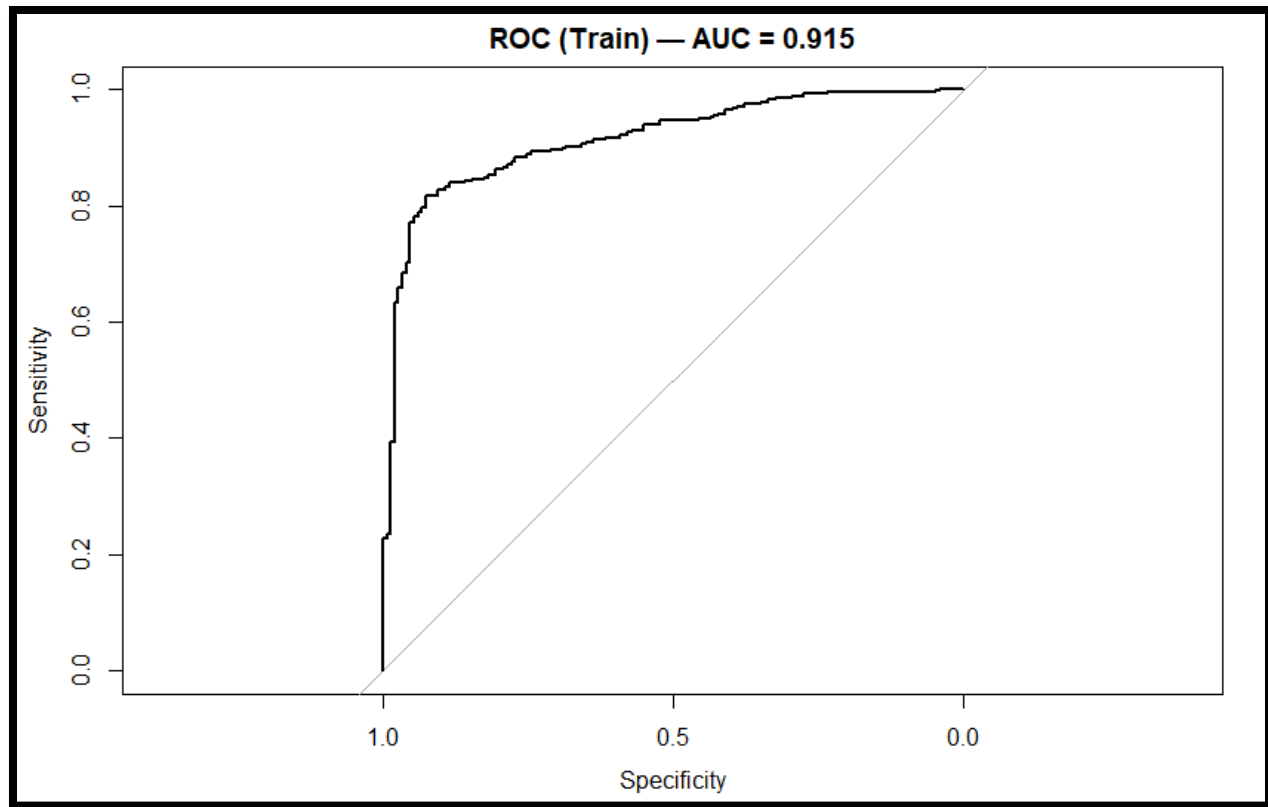
- **Accuracy:** 94.40%
- **Precision:** 97.0%
- **Recall (Sensitivity):** 95.24%
- **Specificity:** 92.19%

This performance is excellent and only slightly lower than the training set performance, which suggests that the model generalizes well to new data and is not significantly overfit.

ROC Curve and AUC

The Receiver Operating Characteristic (ROC) curve for the test set was plotted to visualize the model's performance across all classification thresholds. The curve showed a very strong bow towards the top-left corner, indicating high discriminatory power.

Module 3 Assignment — GLM and Logistic Regression



Module 3 Assignment — GLM and Logistic Regression

The Area Under the Curve (AUC) for the test set was calculated to be **0.921**. This value is very high, confirming that the model has an outstanding ability to distinguish between private and public universities. The close similarity between the training AUC of 0.915 and the test AUC of 0.921 further confirms that the model is robust and not merely memorizing the training data.

Conclusion

This analysis successfully demonstrated that a logistic regression model can be a highly effective tool for classifying universities as public or private. By moving beyond a simple two-variable model and incorporating a more comprehensive set of five predictors (Outstate, Expend, Grad.Rate, S.F.Ratio, and perc.alumni), we constructed a model with outstanding predictive power. The model achieved a test accuracy of 94.4% and an AUC of 0.921, indicating a very high degree of accuracy and an excellent ability to distinguish between the two classes.

The rigorous evaluation on both training and test sets confirmed that the model generalizes well to unseen data, building confidence in its real-world applicability. The findings reinforce widely held understandings of higher education: private universities are strongly characterized by higher tuition, greater per-student spending, more engaged alumni, and lower student-faculty ratios.

This project not only validates the power of generalized linear models for solving classification problems but also provides a clear, data-driven profile of the key factors that define public versus private higher education in the United States. Future work could involve applying more advanced techniques like regularization to potentially simplify the model while maintaining its high performance.

Module 3 Assignment — GLM and Logistic Regression

References

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- R Core Team. (2025). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing.
- Kuhn, M. (2008). Building predictive models in R using the caret package. *Journal of Statistical Software*.
- Robin, X., et al. (2011). pROC: an open-source package to analyze and compare ROC curves. *BMC Bioinformatics*.

Module 3 Assignment — GLM and Logistic Regression

Appendix: R

```
library(ISLR)
library(ggplot2)
library(caret)
library(pROC)
library(dplyr)

set.seed(2025)

data("College")
df <- College

df$Private <- factor(df$Private, levels = c("No", "Yes"))

str(df)
summary(df)

numeric_vars <- c("Apps", "Accept", "Enroll", "Top10perc", "Top25perc", "F.Undergrad",
                  "P.Undergrad", "Outstate", "Room.Board", "Books", "Personal",
                  "PhD", "Terminal", "S.F.Ratio", "perc.alumni", "Expend", "Grad.Rate")

plot_hist <- function(var){
```

Module 3 Assignment — GLM and Logistic Regression

```
ggplot(df, aes(x = .data[[var]])) +  
  geom_histogram(bins = 30) +  
  labs(title = paste("Histogram of", var), x = var, y = "Count")  
}  
  
# Boxplots by Private  
bp1 <- ggplot(df, aes(x = Private, y = Outstate)) + geom_boxplot() + labs(title="Outstate by Private")  
bp2 <- ggplot(df, aes(x = Private, y = Expend)) + geom_boxplot() + labs(title="Expend by Private")  
bp3 <- ggplot(df, aes(x = Private, y = Grad.Rate)) + geom_boxplot() + labs(title="Grad.Rate by Private")  
  
# ---- Train/Test split (70/30) ----  
# Maintain outcome proportions using caret::createDataPartition  
train_index <- createDataPartition(df$Private, p = 0.7, list = FALSE)  
train <- df[train_index, ]  
test <- df[-train_index, ]  
  
# Start with several reasonable predictors known to differ between public/private:  
# Outstate (tuition), Expend (instructional expenditure per student), Grad.Rate, S.F.Ratio, perc.alumni  
  
base_formula <- Private ~ Outstate + Expend + Grad.Rate + S.F.Ratio + perc.alumni  
  
# Fit logistic regression  
fit <- glm(base_formula, data = train, family = binomial(link = "logit"))  
summary(fit)
```

Module 3 Assignment — GLM and Logistic Regression

```
# For simplicity in the script we keep 'fit' as the final model

final_fit <- fit

# Thresholded predictions on TRAIN

train$prob <- predict(final_fit, newdata = train, type = "response")
train$pred <- factor(ifelse(train$prob >= 0.5, "Yes", "No"), levels = c("No", "Yes"))

cm_train <- confusionMatrix(data = train$pred, reference = train$Private, positive = "Yes")
print(cm_train)

# Metrics from caret confusionMatrix

train_accuracy <- cm_train$overall["Accuracy"]
train_precision <- cm_train$byClass["Precision"]
train_recall <- cm_train$byClass["Sensitivity"]
train_specific <- cm_train$byClass["Specificity"]
print(c(Accuracy=train_accuracy, Precision=train_precision, Recall=train_recall, Specificity=train_specific))

# ROC/AUC (TRAIN)

roc_train <- roc(response = train$Private, predictor = train$prob, levels = c("No", "Yes"), direction = "<")
auc_train <- auc(roc_train)
print(auc_train)

# Plot ROC

plot(roc_train, main = sprintf("ROC (Train) — AUC = %.3f", auc_train))

# Thresholded predictions on TEST

test$prob <- predict(final_fit, newdata = test, type = "response")
test$pred <- factor(ifelse(test$prob >= 0.5, "Yes", "No"), levels = c("No", "Yes"))
```

Module 3 Assignment — GLM and Logistic Regression

```
cm_test <- confusionMatrix(data = test$pred, reference = test$Private, positive = "Yes")

print(cm_test)

test_accuracy <- cm_test$overall["Accuracy"]
test_precision <- cm_test$byClass["Precision"]
test_recall <- cm_test$byClass["Sensitivity"]
test_specific <- cm_test$byClass["Specificity"]

print(c(Accuracy=test_accuracy, Precision=test_precision, Recall=test_recall, Specificity=test_specific))

# ROC/AUC (TEST)
roc_test <- roc(response = test$Private, predictor = test$prob, levels = c("No", "Yes"), direction = "<")
auc_test <- auc(roc_test)

print(auc_test)

plot(roc_test, main = sprintf("ROC (Test) — AUC = %.3f", auc_test))
```